The Central Distribution: A Minimax-KL Approach to Noninformative Priors and Feature Selection

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July 11, 2025

Abstract

We introduce the *central distribution*, a probability distribution that represents a set, or more generally a hierarchy, of probability distributions by minimizing the worst-case Kullback–Leibler (KL) divergence to the admissible models. This provides a principled method for converting partial knowledge and set-valued uncertainty into a single probability measure suitable for Bayesian inference.

We show that central distributions admit a minimax characterization and can be obtained as solutions to convex–concave saddle point problems. For parametric models, we establish a sufficient fixed-point condition for priors whose induced marginal distributions are central, revealing a close connection with Bernardo's reference prior methodology. We also extend the framework to hierarchical uncertainty structures and to infinite sequences through the notion of asymptotic central distributions.

Finally, we show how uncertainty about relevant feature subsets can be represented hierarchically, and how the resulting central distributions naturally induce a preference for sparse and structurally simpler models. This provides a unified information-theoretic framework for constructing noninformative priors and performing Bayesian feature selection.

1 Introduction

We begin by revisiting a fundamental decision-making problem and its treatment within Bayesian decision theory.

Consider an environment where an agent must select an action from a set of available actions. The loss associated with each action depends on the unknown state of the environment and can be quantified by a real number. Specifically, let a denote an action, and x represent the unknown environment state. The loss function $\ell(a, x)$ quantifies the loss for each action.

We assume that some evidence e about the unknown environment state is available, and the agent can leverage this evidence to determine the best action. In other words, our agent aims to find an optimal strategy, where a strategy defines the distribution of each action conditioned on the observed evidence: $S(A \mid E)$.

According to Bayesian decision theory, the optimal strategy is the one that minimizes the expected loss:

$$S_{\text{opt}} = \arg\min_S \sum_a \sum_{x,e} \ell(a, x) S(a \mid e) P^*(x, e),$$

where P^* is the joint probability distribution of the environment state and the evidence.

One can show that there always exists a deterministic optimal strategy, and a strategy that for each evidence e selects the action minimizing the expected loss conditioned on e, is optimal:

$$S_{\text{opt}}(e) = \arg\min_a \mathbb{E}_{P^*} \left[\ell(a, x) \mid e \right]. \tag{1}$$

Now, suppose that we decide to use a different evidence E' instead of E. Could this change potentially improve our strategy? Within the framework of Bayesian decision theory, we can easily answer this question: To compare any two strategies, we can simply compare their expected loss. The strategy with the lower expected loss is the better one.

In practice, we cannot choose arbitrary evidence. We can only use evidence that is a deterministic function of the evidence provided by the environment, allowing its value to be determined after observing the original evidence.

Using a simpler form of evidence resembles feature selection in machine learning. For example, if the evidence is an image, we might replace it with the first three Fourier coefficients of the image. Since these coefficients are a function of the original evidence, they satisfy $E' = f(E)$. Under our Bayesian framework, however, the optimal strategy based on E is always at least as good as any strategy based on E'.

One might argue that adopting a strategy based on simpler evidence may discard useful information and is therefore not justified by Bayesian decision theory.

The key point here is that the space of strategies using E' is a subset of the strategies that use E. When we are searching for the best strategy based on E, we are implicitly evaluating all simpler forms as well. This may explain why the standard Bayesian framework does not favor the simpler evidence E'.

With an appropriate choice of prior distribution, the Bayesian framework should automatically extract the relevant information from the evidence when determining an optimal strategy. In this sense, an appropriate prior could induce feature selection. However, when applied in a conventional Bayesian setting, many commonly used noninformative priors, such as the flat prior, do not naturally promote feature selection.

In this paper, we introduce the central distribution, a minimax-KL representative of a set, or more generally a hierarchy, of probability distributions. By establishing a fixed-point condition for priors whose induced marginals are central, this framework offers a new information-theoretic generalization of objective Bayesian reference analysis [3]. It moves beyond maximizing expected missing information to provide a unified mechanism for minimizing worst-case divergence under layered, set-valued uncertainty. Furthermore, we derive asymptotic characterizations of central priors and investigate the role of hierarchical central distributions in Bayesian feature selection.

1.1 Related Work

The problem of constructing objective or noninformative priors has a long history in Bayesian statistics. Among the most influential approaches are Jeffreys priors, reference priors, and maximum-entropy based methods [6, 2, 5].

Jeffreys introduced a general rule for constructing invariant noninformative priors based on the Fisher information matrix [6]. The resulting priors are widely used because of their parameterization invariance and their connection to asymptotic likelihood theory. However, Jeffreys priors are derived from local geometric properties of the parameter space and do not directly address the problem of representing a set of admissible probability distributions by a single distribution.

The closest connection to the present work is Bernardo's reference prior methodology [3, 2]. Reference priors are constructed by maximizing the expected information gained from the data and are intended to represent a state of minimal prior information. In Section 2.3, we show that central priors satisfy a fixed-point condition closely related to the reference prior construction. In particular, for a single parameter and a single set of distributions, the resulting central prior coincides with the corresponding reference prior.

Several authors have also studied objective Bayesian inference through information-theoretic principles [5, 7]. Examples include maximum entropy methods, minimum discrimination information principles, and related divergence-based approaches. These methods typically select a distribution by optimizing an information criterion subject to constraints. In contrast, the central distribution is defined as a minimax representative of a set or hierarchy of distributions, minimizing the worst-case Kullback–Leibler divergence to the admissible models.

The present work is also related to robust Bayesian statistics and imprecise probability [8, 9]. In these frameworks uncertainty is represented by sets of probability distributions rather than a single model. Existing approaches generally focus on performing inference directly with sets of distributions. The central distribution framework instead seeks a single representative distribution that summarizes the available set-valued or hierarchical uncertainty while preserving its information-theoretic structure.

To the best of our knowledge, existing objective-prior methodologies do not provide a unified mechanism for representing hierarchical uncertainty structures and converting them into a single probability distribution through a minimax-KL criterion. The central distribution framework is proposed as a step in that direction.

1.2 Divergence

Suppose that for a given environment, we have specified a probability measure P. In particular, we are interested in the distribution it induces on a random variable X, denoted $P(X)$.

Now, imagine we ask an expert—a person familiar with the environment—to evaluate our choice of P. The expert may hold different beliefs and instead favor another probability measure Q. From the expert's perspective, evaluating P involves assessing the potential loss incurred by using the (in their view) incorrect distribution $P(X)$ when the true distribution is believed to be $Q(X)$.

Let us assume that a real number can be used to measure this loss, and denote it as $\mathbb{D}\left[Q(X) \,\|\, P(X)\right]$. It seems reasonable, as P gets closer to Q this measure decreases, and increases when P diverges from Q. Therefore, $\mathbb{D}$ effectively is a measure of the divergence of $P(X)$ from $Q(X)$, when Q is the believed true probability measure.

Interestingly, if we make certain assumptions about the properties of this divergence measure, it turns out that—up to a constant factor—it must be equal to the Kullback—Leibler (KL) divergence.

Alternatively, by adopting a more minimal set of properties, one can prove that accepting Shannon's entropy as a measure of the uncertainty implies that KL divergence can be considered as an appropriate method for measuring divergence between distributions.

Theorem 1. *Let X be a discrete random variable. Suppose $\mathbb{D}\left[Q(X) \parallel P(X)\right]$ is some divergence measure that quantifies the divergence of $P(X)$ from $Q(X)$. Consider the following properties:*

Consistency with entropy *Let Ω be a continuous random variable, and Q be a probability measure. If $U(\Omega)$ is the uniform distribution with the same support as $Q(\Omega)$, then we have*

$$\mathbb{D}\left[Q(\Omega) \parallel U(\Omega)\right] = \mathbb{H}\left[U(\Omega)\right] - \mathbb{H}\left[Q(\Omega)\right],$$

where $\mathbb{H}$ denotes Shannon's entropy.

Additivity *For any two discrete random variables X, Y and probability measures P, Q*

$$\mathbb{D}\left[Q(X,Y) \parallel P(X,Y)\right] = \mathbb{D}\left[Q(X) \parallel P(X)\right] + \sum_x \mathbb{D}\left[Q(Y|x) \parallel P(Y|x)\right] Q(x)\cdot$$

If $\mathbb{D}\left[\,\cdot\,\right]$ satisfies these properties, then it is the Kullback–Leibler divergence.

Proof. Let $P(X)$ and $Q(X)$ be two arbitrary distributions for a discrete random variable X. We define a new continuous random variable Ω and let

$$p(\Omega = \omega \mid X = x) = \begin{cases} \frac{1}{P(x)} & 0 \leq \omega \leq P(X = x), \\ 0 & \text{otherwise}\cdot \end{cases}$$

Notice that both $P(\Omega|X)$ and $P(\Omega, X)$ are uniform distributions, and

$$\mathbb{H}\left[P(\Omega|x)\right] = \ln P(x),\ \ \mathbb{H}\left[P(\Omega, X)\right] = 0\cdot$$

We define $Q(\Omega|x)$ to be an arbitrary distribution that has the same support as $P(\Omega|x)$. This ensures that $Q(\Omega, X)$ and $P(\Omega, X)$ have the same support as well. Thus, we have

$$\begin{aligned} \mathbb{D}\left[Q(\Omega, X) \parallel P(\Omega, X)\right] &= \mathbb{H}\left[P(\Omega, X)\right] - \mathbb{H}\left[Q(\Omega, X)\right] \\ &= \sum_x \int q(\omega, x) \ln q(\omega, x) d\omega \\ &= \sum_x Q(x) \int q(\omega \mid x) \ln q(\omega, x) d\omega, \end{aligned}$$

and

$$\begin{aligned}\mathbb{D}\left[Q(\Omega|x) \,\|\, P(\Omega|x)\right] &= \mathbb{H}\left[P(\Omega|x)\right] - \mathbb{H}\left[Q(\Omega|x)\right] \\ &= \ln P(x) + \int q(\omega \mid x) \ln q(\omega \mid x) d\omega \cdot\end{aligned}$$

Now, we use the additivity property to obtain a formula for our divergence measure:

$$\begin{aligned}\mathbb{D}\left[Q(X) \,\|\, P(X)\right] &= \mathbb{D}\left[Q(\Omega, X) \,\|\, P(\Omega, X)\right] - \sum_x \mathbb{D}\left[Q(\Omega|x) \,\|\, P(\Omega|x)\right] Q(x) \\ &= \sum_x Q(x) \left(\int q(\omega \mid x) \ln \frac{q(\omega, x)}{q(\omega \mid x)} d\omega - \ln P(x) \right) \\ &= \sum_x Q(x) \left(\ln Q(x) \int q(\omega \mid x) d\omega - \ln P(x) \right) \\ &= \sum_x Q(x) \ln \frac{Q(x)}{P(x)} \cdot \qquad \square\end{aligned}$$

In this theorem, the first property simply states that the entropy of a distribution should be linearly related to the divergence of the uniform distribution from it. The more the divergence is, the less the entropy must be.

The second property tells us that the divergence is an additive quantity. We know that the divergence of $P(X, Y)$ from $Q(X, Y)$ is a measure of the generic loss of using $P(X, Y)$ when we believe $Q(X, Y)$ is the true distribution. To calculate this loss, we can assume that X happens before Y. In that way, this loss would include the loss of using $P(X)$, and then, if we observe $X = x$, which can happen with the probability $Q(x)$, it would also include the loss of using $P(Y \mid X = x)$. This property indicates that for combining these individual terms we should use addition.

We now briefly review some KL divergence properties needed in subsequent sections.

Lemma 1. *Let X be a random variable, and let P, Q, and R be probability distributions over X. The Kullback—Leibler divergence satisfies the following inequality:*

$$\mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, P(X)\right] \geq \sum_x Q(x) \ln \frac{R(x)}{P(x)} \cdot$$

Proof. The Kullback—Leibler divergence is nonnegative and is defined when R is a probability distribution:

$$\mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, R(X)\right] \geq 0 \cdot$$

By adding $\sum_x Q(x) \ln \frac{R(x)}{P(x)}$ to both sides of the above inequality, the desired inequality follows. $\square$

Theorem 2. *Let Q_1, Q_2 and P be probability distributions. If for some $0 \leq \alpha \leq 1$ we have $Q = \alpha Q_1 + (1-\alpha) Q_2$, then:*

$$\mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, P(X)\right] \leq \alpha \mathbb{D}_{\mathrm{KL}}\left[Q_1(X) \,\|\, P(X)\right] + (1-\alpha)\mathbb{D}_{\mathrm{KL}}\left[Q_2(X) \,\|\, P(X)\right] \cdot$$

That is, the KL divergence is convex in its first argument.

Proof. The result follows from Lemma 1 after expanding the left-hand side of the inequality using the definition of KL divergence. $\square$

The KL divergence plays a key role in the notion of central distribution, which we introduce in the next section.

2 Central Distribution

Consider a random variable X. In Bayesian modeling, our knowledge about X is represented by a probability distribution, which serves as a basis for making inferences.

However, in many situations, our information may be too limited to justify choosing a single distribution. Instead, we may be able to specify only a set of plausible distributions. Such a set represents some form of *uncertainty* or *partial knowledge*: although it does not identify a single distribution, it constrains the range of plausible probabilistic models.

A well-known example of this idea is the exchangeability assumption used in de Finetti's theorem. A sequence of random variables $X_1, X_2, \ldots$ is said to be exchangeable if the ordering of the variables carries no information. That is, the joint probability distribution remains invariant under permutations of the variables. This assumption imposes a constraint, effectively introducing a set of plausible distributions. Thus, exchangeability can be seen as a specific form of partial knowledge.

In this work, we adopt the view that a set of probability distributions can represent a meaningful state of partial knowledge, and that such set-valued uncertainty may itself be used as the basis for inference.

More generally, partial knowledge may possess a hierarchical structure. In the simplest case, uncertainty is represented by a single set of candidate distributions. In richer settings, however, this uncertainty may be organized into nested collections, where elements of a set may themselves be sets of distributions. Such nesting can continue recursively, yielding a hierarchical structure.

Crucially, the choice of hierarchical organization is itself informative: grouping distributions into subsets expresses that certain alternatives are regarded as belonging to a common epistemic category and should therefore be aggregated before being compared with alternatives in other groups. Therefore, the hierarchy encodes information beyond the collection of leaf distributions alone.

This hierarchical organization may be interpreted as introducing multiple layers of latent structural decisions, where higher-level choices restrict the admissible lower-level alternatives.

Within a Bayesian framework, inference ultimately requires a single probability measure. Hence, to use hierarchical partial knowledge in Bayesian inference, we require a principled method for transforming a nested set-valued uncertainty into a single probability distribution while preserving the informational structure encoded by the hierarchy.

We adopt the principle that the probability measure used to represent prior beliefs about the environment should depend only on the available information about the environment itself, and not on agent-specific considerations such as the loss function or the informativeness of observations. We call this modeling assumption the *principle of objectivity*.

From a minimax perspective, when uncertainty is represented by a set of candidate distributions, a natural representative distribution is one that minimizes the worst-case divergence to the members of the set. Such a distribution may be viewed as the most conservative or robust single-distribution summary of the available partial knowledge.

Definition 2.1. Let X be a random variable and $\mathcal{Q}$ be a set of probability distributions over X. We call a distribution C a *central distribution* of $\mathcal{Q}$ for X if, for every probability distribution P, we have

$$\sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}} \left[Q(X) \parallel C(X) \right] \leq \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}} \left[Q(X) \parallel P(X) \right], \tag{2}$$

where $\mathbb{D}_{\mathrm{KL}}$ denotes the Kullback—Leibler (KL) divergence.

By convention, any probability distribution is regarded as its own central distribution.

Under this definition, the central distribution of $\mathcal{Q}$ is not necessarily an element of $\mathcal{Q}$. That is, the distribution that best represents the set in the specified sense may lie outside the set it summarizes.

Sometimes partial knowledge is represented by a hierarchical structure of sets of distributions. Our current definition of the central distribution does not account for this case. The following recursive definitions extend the concept to handle such situations. While not directly about feature selection, this extension plays a key role in enabling the connection between central distributions and feature selection, a connection that will become clearer later.

Definition 2.2 (Hierarchy of Distributions)**.** A *hierarchy of distributions* over X is defined recursively as follows:

- Any probability distribution over X is a hierarchy of distributions over X.
- Any set whose elements are hierarchies of distributions over X is itself a hierarchy of distributions over X.

Intuitively, a central distribution acts as a representative or summary of a set of distributions. While it may not belong to the set itself, it reflects the overall properties of the set in a way that is meaningful for inference. It also provides a way to evaluate probability distributions under partial knowledge, motivating the definition of a corresponding central divergence.

The concept of a central distribution is not intended to yield a unique object; rather, all distributions satisfying the definition can be regarded as equally good representatives. However, when multiple central distributions exist, our constructions should not depend on an arbitrary choice among them. To avoid such dependence, we define the central divergence using the best divergence attained over the set of all central distributions.

Definition 2.3 (Central Divergence)**.** For any distribution P, we define the *central divergence* of P from a hierarchy of distributions $\mathcal{H}$, with respect to a random variable X, as follows:

$$\mathbb{D}_{\mathrm{C}}\left[\mathcal{H}(X) \,\|\, P(X)\right] = \inf_{C} \mathbb{D}_{\mathrm{KL}}\left[C(X) \,\|\, P(X)\right],$$

where C denotes a central distribution of $\mathcal{H}$ for X.

So far, this definition applies only when the hierarchy is a single set of distributions. The following definition extends the construction recursively to general hierarchical uncertainty structures.

Definition 2.4 (Central Distribution)**.** Let X be a random variable, and let $\mathcal{H}^*$ be a hierarchy of distributions over X. We call a distribution C a *central distribution* of $\mathcal{H}^*$ if, for every probability distribution P, we have:

$$\sup_{\mathcal{H}\in\mathcal{H}^*} \mathbb{D}_{\mathrm{C}}\left[\mathcal{H}(X) \,\|\, C(X)\right] \leq \sup_{\mathcal{H}\in\mathcal{H}^*} \mathbb{D}_{\mathrm{C}}\left[\mathcal{H}(X) \,\|\, P(X)\right].$$

Definition 2.4 generalizes Definition 2.1, which is recovered when the hierarchy consists of a set of probability distributions.

For later use, we introduce a compact notation for the minimax objective appearing in Definition 2.1.

Notation 2.1. For a set of probability distributions $\mathcal{Q}$ over X, we define the functional

$$\mathbb{S}_{\mathcal{Q}}\left[P(X)\right] \coloneqq \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, P(X)\right].$$

When $\mathcal{Q}$ and X are clear from the context, we may simply write $\mathbb{S}\left[P\right]$.

This notation will be used throughout the sequel and allows Definition 2.1 to be written compactly as

$$C \in \arg\min_{P} \mathbb{S}_{\mathcal{Q}}\left[P(X)\right].$$

A simple example illustrates that hierarchical organization can affect the resulting representative distribution even when the underlying admissible sets remain unchanged.

Let X be a random variable with three possible outcomes x_1, x_2, x_3. For each i, let S_i denote the set of all probability distributions over X that assign zero probability to x_i.

By symmetry, the central distribution of each set S_i is the uniform distribution over its support:

$$C_{S_1} = (0, 0.5, 0.5)\,, \qquad C_{S_2} = (0.5, 0, 0.5)\,, \qquad C_{S_3} = (0.5, 0.5, 0)\,.$$

If the three sets S_1, S_2, S_3 are treated symmetrically as elements of a single set, then their central distribution is

$$C_{\{S_1,S_2,S_3\}} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right).$$

Now consider instead a hierarchical organization in which S_1 and S_2 are grouped together, while S_3 forms its own separate branch. By numerically solving the minimax problem in Definition 2.4, we obtain

$$C_{\{S_1,S_2\}} = (0.25, 0.25, 0.5)\,.$$

Applying the same procedure to the two distributions $C_{\{S_1,S_2\}}$ and C_{S_3} yields

$$C_{\{\{S_1,S_2\},S_3\}} = (0.4, 0.4, 0.2)\,.$$

Thus, although the underlying admissible sets are identical in both constructions, the resulting representative distribution differs.

This demonstrates that hierarchical structure carries informational content beyond the collection of leaf distributions alone. Consequently, flattening a hierarchy into a single unstructured set may discard meaningful epistemic structure.

In many learning problems, uncertainty concerns not only the parameters of a predictive model but also its structural form. In particular, when the relevance of potential features is unknown, each candidate feature subset determines a distinct set of plausible models in which the target variable depends only on that feature subset.

Feature selection may therefore be viewed as uncertainty over a collection of distributions. The hierarchical central distribution framework provides a natural mechanism for representing and aggregating such uncertainty. By organizing candidate models hierarchically, one can encode preferences over structural hypotheses, such as favoring sparse subsets, grouping related attributes, or privileging particular interactions.

Candidate feature subsets typically induce overlapping families of admissible distributions. Distributions representing simpler dependency structure may belong simultaneously to many such families, whereas more complex dependency patterns belong to fewer. As a result, hierarchical aggregation through central distributions can induce a preference toward structurally simpler models by assigning more weight to distributions compatible with multiple structural hypotheses.

2.1 Minimax Characterization

With the concept of hierarchical central distributions in place, we now develop the theoretical machinery needed to characterize and compute them. The following results provide optimization-theoretic characterizations of central distributions that form the foundation of the subsequent analysis.

Throughout this section, all random variables are assumed to be discrete with finite support. Before proceeding, we establish the existence of central distributions in the finite setting considered throughout this section.

Theorem 3. *Let X be a discrete random variable with finite support, and let $\mathcal{Q}$ be a finite set of probability distributions over X. Then a central distribution of $\mathcal{Q}$ for X exists.*

Proof. For a fixed distribution Q, we may write

$$\mathbb{D}_{\mathrm{KL}}\left[Q(X) \parallel P(X)\right] = c - \sum_{x} Q(x) \ln P(x).$$

Since ln is upper semicontinuous on $[0, +\infty)$ and X has finite support, it follows that $\mathbb{D}_{\mathrm{KL}}\left[Q(X) \parallel P(X)\right]$ is lower semicontinuous as a function of P.

Because $\mathcal{Q}$ is finite, $\mathbb{S}_{\mathcal{Q}}\left[P(X)\right]$ is the maximum of finitely many lower semicontinuous functions and is therefore itself lower semicontinuous.

The set of probability distributions over X is a compact simplex. Hence, by the extreme value theorem, $\mathbb{S}_{\mathcal{Q}}$ attains its minimum. Therefore there exists a probability distribution C such that

$$\mathbb{S}_{\mathcal{Q}}\left[C(X)\right] = \inf_{P} \mathbb{S}_{\mathcal{Q}}\left[P(X)\right].$$

Thus C is a central distribution of $\mathcal{Q}$ for X. □

The central distribution is essentially the solution to a convex–concave game. Because solutions of convex–concave games correspond to saddle points, by defining the appropriate objective function, we can characterize the central distribution using standard convex optimization techniques.

Definition 2.5 (Saddle Point)**.** Consider a function $f : \mathcal{W} \times \mathcal{Z} \to \mathbb{R}$. We refer to a pair (w^*, z^*) as a *saddle point* for f, if we have

$$f(w^*, z) \leq f(w^*, z^*) \leq f(w, z^*),$$

for all $w \in \mathcal{W}$ and $z \in \mathcal{Z}$.

Next, we establish the connection between saddle points and central distributions. One significant insight provided by the next theorem is that it suggests inference and searching for a central distribution can be combined and performed simultaneously when finding a saddle point.

Theorem 4. *Let X be a discrete random variable, and $\mathcal{Q}$ be a finite set of distributions over X. Consider the following function:*

$$f(P, R) = \sum_{i} \mathbb{D}_{\mathrm{KL}}\left[Q_i(X) \parallel P(X)\right] R(i),$$

where P and R are probability distributions, $Q_i \in \mathcal{Q}$ and the index i ranges over all elements of $\mathcal{Q}$.

If (P^*, R^*) is a saddle point for f, then P^* is a central distribution of $\mathcal{Q}$ for X.

Proof. Let C be a central distribution of $\mathcal{Q}$ for X, and (P^*, R^*) be a saddle point for f. Since X is a discrete random variable and $\mathcal{Q}$ is finite, C exists. As C is a probability distribution, the point (C, R^*) lies within the domain of f. By the definition of a saddle point, this implies that

$$f(P^*, R^*) \leq f(C, R^*). \tag{3}$$

Because R^* is a probability distribution, the definition of f implies:

$$f(C, R^*) \leq \mathbb{S}\left[C\right].$$

On the other hand, by the definition of a saddle point

$$f(P^*, R^*) = \max_R f(P^*, R),$$

where R is a probability distribution, and therefore

$$\max_R f(P^*, R) = \mathbb{S}\left[P^*\right].$$

Thus, Equation 3 implies

$$\mathbb{S}\left[P^*\right] \leq \mathbb{S}\left[C\right].$$

Since C is a central distribution, we must have

$$\mathbb{S}\left[P^*\right] = \mathbb{S}\left[C\right],$$

and P^* is also a central distribution of $\mathcal{Q}$ for X. □

Lemma 2. *Let f be the function defined in Theorem 4. The point (P^*, R^*) is a saddle point for f, if and only if there are constants λ, η and a function μ, such that for every i, x the following conditions hold:*

$$\sum_x Q_i(x) \ln \frac{Q_i(x)}{P^*(x)} - \mu(i) = \lambda, \tag{4}$$

$$P^*(x) = \sum_i R^*(i) Q_i(x) \tag{5}$$

$$\sum_i R^*(i) = 1, \tag{6}$$

$$R^*(i) \geq 0, \tag{7}$$

$$\mu(i) \geq 0,$$

$$\mu(i) R^*(i) = 0. \tag{8}$$

Furthermore, for a fixed probability distribution R, a distribution $\hat{P}$ minimizes $f(\,\cdot\,, R)$, if and only if for all x it satisfies

$$\hat{P}(x) = \sum_i Q_i(x) R(i). \tag{9}$$

Proof. Similar to f, we define $\hat{f}$ as follows:

$$\hat{f}(P, R) = \sum_i R(i) \sum_x Q_i(x) \ln \frac{Q_i(x)}{P(x)},$$

where the constraints that P and R be probability distributions are omitted. This enables us to formulate the saddle point of f as the solution to a constrained optimization problem defined for $\hat{f}$.

Let (P^*, R^*) be a simultaneous solution to the following two optimization problems:

$$\begin{array}{lll} \text{minimize} & \hat{f}(P, R) & \text{w.r.t. } P \\ \text{subject to} & \sum_x P(x) = 1, & \\ & & \\ \text{maximize} & \hat{f}(P, R) & \text{w.r.t. } R \\ \text{subject to} & R(i) \geq 0, & \text{for all } i \\ & \sum_i R(i) = 1 & \end{array} \tag{10}$$

Clearly, if (P^*, R^*) exists, it will be a saddle point for f.

Both of these optimization problems are convex and differentiable in the sense of convex optimization; see [4]. To establish this, consider that all the constraint functions are affine, and therefore are convex, concave and differentiable.

Additionally, $\hat{f}$ is a differentiable function with respect to both R and P. For any fixed P, $\hat{f}(P, \cdot)$ is an affine function and is concave.

On the other hand, for a fixed solution to the second optimization problem denoted by $\hat{R}$, we can write

$$\hat{f}(P, \hat{R}) = c - \sum_{i,x} \hat{R}(i) Q_i(x) \ln P(x), \tag{11}$$

where c is not a function of P. Since $\hat{R}$ is a solution to the second optimization problem, it satisfies the corresponding constraints and is therefore a valid probability distribution. Hence, for all i, x we have $\hat{R}(i)Q_i(x) \geq 0$, and $\hat{f}(\cdot, \hat{R})$ is the negative of a linear combination of concave $\ln P(x)$ functions with nonnegative coefficients. This function is convex.

Thus, both optimization problems are convex and differentiable, with affine constraints. The constraints simply restrict the feasible set to the set of probability distributions, and it is always possible to find feasible points that meet all the constraints. Therefore, Slater's condition is satisfied, and the KKT conditions provide necessary and sufficient conditions for optimality [4]. In other words, if there exists a point (P^*, R^*) that satisfies the KKT conditions for both problems, it will be a saddle point for f, and vice versa.

The KKT conditions are as follows for every x, i and constants λ, η:

$$\sum_i R^*(i) Q_i(x) = \eta P^*(x) \tag{12}$$

$$\sum_x P^*(x) = 1, \tag{13}$$

$$\sum_x Q_i(x) \ln \frac{Q_i(x)}{P^*(x)} - \mu(i) = \lambda,$$

$$\sum_i R^*(i) = 1, \tag{14}$$

$$R^*(i) \geq 0,$$

$$\mu(i) \geq 0,$$

$$\mu(i) R^*(i) = 0.$$

By summing Equation 12 over x and using Equation 14, we have

$$\eta = 1\,.$$

Thus, we get Equation 5 which also implicitly implies Equation 13.

To prove the second part of the lemma, we observe that when R is a probability distribution, Equations 12 and 13 are the KKT conditions for the minimization problem in Equation 10. Since this minimization problem is convex and satisfies Slater's condition, the KKT conditions are necessary and sufficient. Hence

$$\hat{P}(x) = \sum_i Q_i(x) R(i)$$

if and only if $\hat{P}$ minimizes $f(\,\cdot\,, R)$, where R is any fixed distribution. □

Lemma 3. *Let $\mathcal{Q}$ be a finite set of probability distributions over a discrete random variable X. Define the function g as:*

$$g(R) = \sum_i \mathbb{D}_{\mathrm{KL}} \left[Q_i(X) \parallel R(X) \right] R(i),$$

where $Q_i \in \mathcal{Q}$, and R is a probability distribution such that

$$R(X, I) = Q_i(X) R(I).$$

Let f be the function defined in Theorem 4. If (P^*, R^*) is a saddle point for f, then the joint distribution $Q_i(X) R^*(I)$ is a maximizer of g.

Proof. Let (P^*, R^*) be a saddle point for f. By Lemma 2, P^* satisfies Equation 5. Letting $R^*(X, I) = Q_i(X)R^*(I)$ we have

$$g(R^*) = f(P^*, R^*).$$

Now, let R be an arbitrary probability distribution such that $g(R)$ is defined. From the second part of Lemma 2, it follows that $g(R)$ indeed corresponds to the minimum of $f(\,\cdot\,, R)$, which implies

$$g(R) \leq f(P^*, R).$$

On the other hand, since (P^*, R^*) is a saddle point, we also have

$$f(P^*, R) \leq f(P^*, R^*).$$

Combining the two inequalities, we get:

$$g(R) \leq g(R^*).$$

This shows that $g(R^*)$ is the maximum of g, as desired. $\square$

The following theorem gives a standard sufficient condition for optimality in the KL-minimax setting which is the rationale behind the term "central distribution." Like the center of a circle, the central distribution is equidistant from the boundary.

Theorem 5 (Centrality)**.** *Let X be a discrete random variable, and let $\mathcal{Q}$ be a finite set of distributions over X. Suppose C is a convex combination of elements of $\mathcal{Q}$, with all mixture weights strictly positive. If there exists a constant λ such that*

$$\mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, C(X)\right] = \lambda, \tag{15}$$

for every $Q \in \mathcal{Q}$, then C is the unique central distribution of $\mathcal{Q}$ for X.

Proof. Lemma 2 proves a set of sufficient conditions for a saddle point. Note that more restrictive conditions can also be imposed to derive an alternative set of sufficient conditions. Suppose additionally that $R^*(i) > 0$ for every i. Then Equation 8 implies $\mu(i) = 0$, and Lemma 2 conditions can be simplified to

$$\sum_x Q(x) \ln \frac{Q(x)}{P^*(x)} = \lambda,$$
$$P^*(x) = \sum_i R^*(i) Q_i(x),$$

which must hold for every $Q \in \mathcal{Q}$ and some strictly positive probability distribution R^*. Since C is a convex combination of elements of $\mathcal{Q}$, the corresponding convex weights define the distribution R^* and C satisfies these conditions.

Therefore, by Lemma 2 and Theorem 4, C is a central distribution of $\mathcal{Q}$ for X.

On the other hand, for any probability distribution C', we have

$$\begin{aligned}
\mathbb{D}_{\mathrm{KL}}\left[C(X) \,\|\, C'(X)\right] &= \sum_i R^*(i) \left(\mathbb{D}_{\mathrm{KL}}\left[Q_i(X) \,\|\, C'(X)\right] - \mathbb{D}_{\mathrm{KL}}\left[Q_i(X) \,\|\, C(X)\right]\right) \\
&= \sum_i R^*(i) \mathbb{D}_{\mathrm{KL}}\left[Q_i(X) \,\|\, C'(X)\right] - \lambda \\
&\leq \mathbb{S}\left[C'\right] - \lambda. \qquad (16)
\end{aligned}$$

If C' is a central distribution of $\mathcal{Q}$ for X, the definition of a central distribution implies

$$\mathbb{S}\left[C'\right] = \mathbb{S}\left[C\right] = \lambda.$$

Therefore, by non-negativity of the KL divergence, we have

$$\mathbb{D}_{\mathrm{KL}}\left[C(X) \,\|\, C'(X)\right] = 0.$$

Hence C and C' are identical probability distributions. $\square$

Corollary 5.1. *Under the assumptions of Theorem 5, C is a maximizer of the function g defined in Lemma 3.*

Proof. In the proof of Theorem 5, we showed that (C, R^*) is a saddle point for f. By letting $C(X, I) = Q_i(X)R^*(I)$, Lemma 3 implies that C is a maximizer of the function g. $\square$

Lemma 4. *Assuming the conditions of Theorem 5 and using the same λ, if P is a convex combination of elements of $\mathcal{Q}$, then there exists some $Q_0 \in \mathcal{Q}$ such that*

$$\mathbb{D}_{\mathrm{KL}}\left[Q_0(X) \,\|\, P(X)\right] \leq \lambda.$$

Proof. We proceed by contradiction. Suppose that for every $Q \in \mathcal{Q}$, we have $\mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, P(X)\right] > \lambda$. This implies that

$$g(P) > \lambda = g(C),$$

where C is the distribution specified in Theorem 5, and g is the function defined in Lemma 3. Note that $g(P)$ is defined, because P is a convex combination of elements of $\mathcal{Q}$.

However, by Corollary 5.1, the distribution C maximizes g, which contradicts the inequality $g(P) > g(C)$. $\square$

2.2 Approximate Central Distributions

Suppose we want to approximate the central distribution of a hierarchy of distributions $\mathcal{H}$. To do so, we need a measure that quantifies the quality of an approximation. We argue that a natural choice is the central divergence introduced in Definition 2.3. Since central distributions are intended to serve as representatives of hierarchies of distributions, central divergence may be viewed as a measure of how well a probability distribution represents a hierarchy.

At first glance, computing this quantity appears to require explicit knowledge of the central distribution of $\mathcal{H}$. However, it turns out that the central divergence admits an upper bound that can be estimated without explicitly knowing the central distribution.

In variational settings, it is often more convenient to work with unnormalized measures rather than normalized distributions. Therefore, we state the next theorem in terms of unnormalized measures. We assume throughout that $\tilde{P}(x) > 0$ whenever $Q(x) > 0$ for some $Q \in \mathcal{Q}$, so that all logarithms and KL divergences are well-defined.

Theorem 6. *Let $\mathcal{Q}$ be a finite set of probability distributions over a discrete random variable X, and suppose the assumptions of Theorem 5 hold. Let $\tilde{P}$ be a nonnegative linear combination of elements of $\mathcal{Q}$, and let P be the corresponding normalized distribution. Then*

$$\mathbb{D}_{\mathrm{C}}\left[\mathcal{Q}(X) \,\|\, P(X)\right] \leq \sup_{Q\in\mathcal{Q}} \mathbb{F}\left[Q, \tilde{P}\right] - \inf_{Q\in\mathcal{Q}} \mathbb{F}\left[Q, \tilde{P}\right],$$

where

$$\mathbb{F}\left[Q, \tilde{P}\right] = \mathbb{H}\left[Q(X)\right] + \mathbb{E}_Q\left[\ln \tilde{P}(X)\right].$$

Proof. By Theorem 5, C is the unique central distribution of $\mathcal{Q}$, and therefore from Equation 16, we have

$$\mathbb{D}_{\mathrm{C}}\left[\mathcal{Q}(X) \,\|\, P(X)\right] \leq \mathbb{S}\left[P\right] - \lambda.$$

On the other hand, since P is a convex combination of elements of $\mathcal{Q}$, Lemma 4 implies that

$$\inf_{Q\in\mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, P(X)\right] \leq \lambda.$$

Combining these two inequalities yields

$$\mathbb{D}_{\mathrm{C}}\left[\mathcal{Q}(X) \,\|\, P(X)\right] \leq \mathbb{S}\left[P\right] - \inf_{Q\in\mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, P(X)\right]. \tag{17}$$

Now let $\tilde{P}(X) = ZP(X)$, where $Z > 0$ is a normalization constant. Then, for any Q,

$$\mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, P(X)\right] = \ln Z - \mathbb{F}\left[Q, \tilde{P}\right].$$

Since $\ln Z$ does not depend on Q, the supremum and infimum translate directly into the corresponding expressions in terms of $\mathbb{F}$, and the result follows from Equation 17. $\square$

The upper bound in Equation 17 becomes small precisely when the variation of $\mathbb{F}$ across $Q \in \mathcal{Q}$ is small. The theorem therefore suggests seeking a $\tilde{P}$ such that $\mathbb{F}\left[Q, \tilde{P}\right]$ is approximately constant over $Q \in \mathcal{Q}$. This constant can be arbitrary, but it is convenient to choose it to be zero, in which case

$$\mathbb{E}_Q\left[\ln \tilde{P}(X)\right] \approx -\mathbb{H}\left[Q(X)\right].$$

Note that the bound in Theorem 6 is guaranteed only when $\tilde{P}$ is a nonnegative linear combination of elements of $\mathcal{Q}$. Furthermore, there must exist $\tilde{C}$, a strictly positive linear combination of elements of $\mathcal{Q}$, such that for every $Q \in \mathcal{Q}$ the equality holds:

$$\mathbb{E}_Q\left[\ln \tilde{C}(X)\right] = -\mathbb{H}\left[Q(X)\right].$$

2.3 Parametric Families and Central Priors

The results of the previous subsection were established under the assumption that $\mathcal{Q}$ is finite. Many of the underlying constructions, however, naturally extend to parametric families where $\mathcal{Q}$ can be indexed by a finite-dimensional real vector Θ:

$$\mathcal{Q} = \{Q_\theta \mid \theta \in \Theta\}.$$

We do not pursue the most general extension here. Instead, we use this parametric formulation to motivate the notion of a central prior and to study its relationship with central distributions. We can treat Θ as a random variable and interpret Q_θ as a conditional distribution $Q(x \mid \theta)$. In this setting, the weight function $R(i)$ appearing in the finite case becomes a prior distribution over Θ.

Definition 2.6 (Central Prior)**.** Let X be a discrete random variable, and let $\mathcal{Q}$ be a set of probability distributions over X, indexed by a parameter Θ. A nonnegative measure c on Θ is called a central prior of $\mathcal{Q}$ for X if,

$$C(x) = \int Q(x \mid \theta) c(\theta) d\theta$$

is a central distribution of $\mathcal{Q}$ for X.

Definition 2.7 (Interior Distribution)**.** Let $\mathcal{Q}$ be a set of probability distributions over a random variable X. A distribution R is called an *interior distribution* of $\mathcal{Q}$ if, for every probability distribution P, there exists some $Q \in \mathcal{Q}$ such that

$$\mathbb{D}_{\mathrm{KL}}\left[R(X) \,\|\, P(X)\right] \leq \mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, P(X)\right].$$

Theorem 7. *Any distribution that can be expressed as a convex combination of elements of $\mathcal{Q}$ is an interior distribution of $\mathcal{Q}$.*

Proof. Let $R = \sum_i \alpha_i Q_i$, where $Q_i \in \mathcal{Q}$, $\alpha_i \geq 0$, and $\sum_i \alpha_i = 1$.

Fix an arbitrary probability distribution P. By the convexity of KL divergence in its first argument, we have:

$$\mathbb{D}_{\mathrm{KL}}\left[R(X) \,\|\, P(X)\right] \leq \sum_i \alpha_i \mathbb{D}_{\mathrm{KL}}\left[Q_i(X) \,\|\, P(X)\right].$$

Now, note that the right-hand side is a weighted average of the KL divergences. So, there must be at least one index j such that

$$\mathbb{D}_{\mathrm{KL}}\left[R(X) \,\|\, P(X)\right] \leq \mathbb{D}_{\mathrm{KL}}\left[Q_j(X) \,\|\, P(X)\right].$$

Since $Q_j \in \mathcal{Q}$, the condition in Definition 2.7 is satisfied, and R is indeed an interior distribution of $\mathcal{Q}$. $\square$

Theorem 8. *Let R be an interior distribution of $\mathcal{Q}$. If C is a central distribution of $\mathcal{Q}$, it is also a central distribution of $\mathcal{Q} \cup \{R\}$ and vice versa.*

Proof. This theorem follows directly from the definition of an interior distribution, which implies that for any probability distribution P, we have

$$\mathbb{S}_{\mathcal{Q}}\left[P\right] = \mathbb{S}_{\mathcal{Q}\cup\{R\}}\left[P\right]. \qquad \square$$

One important consequence of this theorem is that, in order to identify a central distribution within a parametric model, it suffices to restrict attention to the set of likelihood distributions induced by different values of the model parameters.

The next theorem is a standard result that is typically used in the derivation of central distributions.

Theorem 9. *Let X and T be two random variables, and let $\mathcal{Q}$ be a set of distributions over the pair (X, T). Assume that for every $Q \in \mathcal{Q}$, we have*

$$Q(X, T) = R(X \mid T)Q(T),$$

where R is a fixed conditional distribution. If C is a central distribution of $\mathcal{Q}$ for T, then $R(X \mid T)C(T)$ is a central distribution of $\mathcal{Q}$ for (X, T).

Proof. Let C be a central distribution of $\mathcal{Q}$ for T. For any arbitrary probability distribution P, the definition of a central distribution implies

$$\mathbb{S}\left[R(X \mid T)C(T)\right] \leq \mathbb{S}\left[R(X \mid T)P(T)\right]. \tag{18}$$

Since the KL divergence is nonnegative, for every probability distribution P and every $Q \in \mathcal{Q}$ we have

$$\sum_t \mathbb{D}_{\mathrm{KL}}\left[R(X \mid t) \,\|\, P(X \mid t)\right] Q(t) \geq 0.$$

That implies

$$\mathbb{D}_{\mathrm{KL}}\left[Q(X,T) \,\|\, R(X \mid T)P(T)\right] \leq \mathbb{D}_{\mathrm{KL}}\left[Q(X,T) \,\|\, P(X,T)\right].$$

Taking the supremum over $Q \in \mathcal{Q}$ preserves the inequality, since it holds pointwise for every $Q \in \mathcal{Q}$:

$$\mathbb{S}\left[R(X \mid T)P(T)\right] \leq \mathbb{S}\left[P(X,T)\right].$$

Combining this inequality with Equation 18, we conclude that $R(X \mid T)C(T)$ is a central distribution of $\mathcal{Q}$ for (X,T). $\square$

The next theorem highlights a formal connection between the central distribution framework and Bernardo's reference prior methodology [3].

Theorem 10. *Let X be a discrete random variable, and let $\mathcal{Q}$ be a set of probability distributions over X, indexed by a finite-dimensional vector Θ. Let c be a strictly positive prior distribution for Θ such that for all $Q(x \mid \theta) \in \mathcal{Q}$ we have*

$$c(\theta) \propto \exp\left\{\sum_x Q(x \mid \theta) \ln c(\theta \mid x)\right\}. \tag{19}$$

Then, c is a central prior of $\mathcal{Q}$ for X.

Proof. From Equation 19, it follows that there exists a constant λ such that

$$c(\theta) = \exp\left\{-\lambda + \sum_x Q(x \mid \theta) \ln \frac{Q(x \mid \theta)c(\theta)}{C(x)}\right\}.$$

Since $c(\theta)$ is strictly positive,

$$\begin{aligned} c(\theta) &= \exp\left\{-\lambda + \mathbb{D}_{\mathrm{KL}}\left[Q_\theta(X) \,\|\, C(X)\right] + \ln c(\theta)\right\} \\ &= c(\theta) \exp\left\{-\lambda + \mathbb{D}_{\mathrm{KL}}\left[Q_\theta(X) \,\|\, C(X)\right]\right\} \end{aligned}$$

implies

$$\mathbb{D}_{\mathrm{KL}}\left[Q_\theta(X) \,\|\, C(X)\right] = \lambda\,.$$

This holds for all θ, and c is a strictly positive distribution, by Theorem 5 it follows that C is a central distribution of $\mathcal{Q}$ for X. Thus, by definition, c is a central prior of $\mathcal{Q}$ for X. $\square$

This theorem provides a sufficient fixed-point characterization of central priors; it does not address existence or uniqueness in full generality.

Although this theorem is proved directly from Theorem 5, Lemma 3 suggests an alternative route to the connection with Bernardo's reference prior methodology. Since reference priors are defined by maximizing missing information, the optimization characterization provided by Lemma 3 also offers another perspective on this relationship.

2.4 Asymptotic Central Distributions

Many probabilistic models are naturally formulated in terms of a sequence of random variables whose length is not fixed in advance. For example, when a coin is tossed repeatedly, there is no natural upper bound on the number of tosses, and any finite sample can be viewed as an initial segment of an infinite sequence.

Motivated by such examples, let S_n denote the first n variables of an infinite sequence of random variables:

$$S_n = (X_1, X_2, \ldots, X_n).$$

The collection $\{S_n\}_{n=1}^{\infty}$ therefore forms a sequence of random variables.

In many settings, our uncertainty about S_n can be represented by a central distribution. A prominent example arises when the sequence is assumed to be exchangeable. As discussed earlier, exchangeability can be viewed as a form of partial knowledge that defines a set of plausible distributions, from which a central distribution for S_n can be constructed.

However, this central distribution generally depends on the sequence length n. According to the principle of objectivity, our beliefs should depend only on the available information and not on arbitrary aspects of the experimental setup, such as the number of variables under consideration.

Importantly, a distribution defined over a longer sequence induces marginal distributions over all shorter initial segments. This suggests that, rather than constructing a separate central distribution for each S_n, we should seek a single probability measure over the infinite sequence, from which the distributions of all finite initial segments can be obtained as marginals.

This motivates the construction of a central distribution that is well defined over S_∞.

Definition 2.8. Let $\{S_n\}_{n=1}^{\infty}$ be a sequence of random variables and $\mathcal{Q}$ be a set of probability measures, where each element of $\mathcal{Q}$ defines a distribution over each S_n. Assume C_n is a central distribution for S_n. Let C be a probability measure that defines a distribution over every S_n. C is a central distribution of $\mathcal{Q}$ for $\{S_n\}_{n=1}^{\infty}$, if we have

$$\lim_{n\to\infty} \mathbb{D}_{\mathrm{KL}}\left[C_n(S_n) \,\|\, C(S_n)\right] = 0.$$

Intuitively, this definition states that as $n \to \infty$ the distribution that C induces over S_n becomes increasingly similar to a central distribution for S_n. In other words, C is a central distribution for the infinite sequence S_∞.

Our next goal is to extend Theorem 5 so that it also applies in this asymptotic setting, thereby providing a characterization of central distributions for infinite sequences.

Theorem 11 (Asymptotic Centrality)**.** *Let $\{S_n\}_{n=1}^{\infty}$ be a sequence of random variables and $\mathcal{Q}$ be a set of probability measures, where each element of $\mathcal{Q}$ defines a distribution over each S_n. Assume that there exists a natural number M such that, for every $n \geq M$, there exists a distribution C_n satisfying the conditions of Theorem 5 for S_n.*

Let C be a probability measure that defines a distribution over every S_n. If there exists a sequence of constants $\{\lambda_n\}_{n=1}^{\infty}$ such that

$$\lim_{n\to\infty} \sup_{Q\in\mathcal{Q}} \left|\mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C(S_n)\right] - \lambda_n\right| = 0,$$

then C is a central distribution of $\mathcal{Q}$ for $\{S_n\}_{n=1}^{\infty}$.

Proof. For every $n \geq M$, by Theorem 5, there exists a constant μ_n such that for all $Q \in \mathcal{Q}$

$$\mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C_n(S_n)\right] = \mu_n.$$

By the definition of a central distribution we have

$$\mathbb{S}\left[C(S_n)\right] \geq \mu_n,$$

and, by Lemma 4, it also follows that

$$\inf_{Q\in\mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C(S_n)\right] \leq \mu_n.$$

On the other hand, clearly

$$\left|\mathbb{S}\left[C(S_n)\right] - \lambda_n\right| \leq \sup_{Q\in\mathcal{Q}} \left|\mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C(S_n)\right] - \lambda_n\right|,$$

and,

$$\left| \inf_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}} \left[Q(S_n) \,\|\, C(S_n) \right] - \lambda_n \right| \leq \sup_{Q \in \mathcal{Q}} \left| \mathbb{D}_{\mathrm{KL}} \left[Q(S_n) \,\|\, C(S_n) \right] - \lambda_n \right|.$$

Consequently, by the squeeze theorem, the sequence $\{\mu_n - \lambda_n\}_{n=1}^{\infty}$ must converge to 0, and for any arbitrary $Q \in \mathcal{Q}$, we have

$$\begin{aligned}
\lim_{n\to\infty} \sum_{s_n} Q(s_n) \ln \frac{C_n(s_n)}{C(s_n)} &= \lim_{n\to\infty} \left(\mathbb{D}_{\mathrm{KL}} \left[Q(S_n) \,\|\, C(S_n) \right] - \mathbb{D}_{\mathrm{KL}} \left[Q(S_n) \,\|\, C_n(S_n) \right] \right) \\
&= \lim_{n\to\infty} \left(\mathbb{D}_{\mathrm{KL}} \left[Q(S_n) \,\|\, C(S_n) \right] - \mu_n + \lambda_n - \lambda_n \right) \\
&= \lim_{n\to\infty} \left(\mathbb{D}_{\mathrm{KL}} \left[Q(S_n) \,\|\, C(S_n) \right] - \lambda_n \right) - \lim_{n\to\infty} (\mu_n - \lambda_n) \\
&= 0.
\end{aligned}$$

Since C_n satisfies the conditions of Theorem 5, it is a convex combination of elements of $\mathcal{Q}$. Therefore, $\mathbb{D}_{\mathrm{KL}} \left[C_n(S_n) \,\|\, C(S_n) \right]$ can be expressed as a convex combination of the quantities

$$Q_i(s_n) \ln \frac{C_n(s_n)}{C(s_n)},$$

where $Q_i \in \mathcal{Q}$.

By the previous result, we can conclude

$$\lim_{n\to\infty} \mathbb{D}_{\mathrm{KL}} \left[C_n(S_n) \,\|\, C(S_n) \right] = 0.$$

Thus, by Definition 2.8, C is a central distribution of $\mathcal{Q}$ for the sequence $\{S_n\}_{n=1}^{\infty}$. □

We now present an asymptotic version of Theorem 9, applicable to infinite sequences of random variables.

Theorem 12. *Let $\{S_n\}_{n=1}^{\infty}$ and $\{T_n\}_{n=1}^{\infty}$ be two sequences of random variables. Let $\mathcal{Q}$ be a set of probability measures, where each $Q \in \mathcal{Q}$ defines a joint distribution over (S_n, T_n) for every n.*

Assume there exists a natural number M and a fixed conditional distribution R such that, for all $Q \in \mathcal{Q}$ and all $n \geq M$,

$$Q(S_n, T_n) = R(S_n \mid T_n) Q(T_n).$$

Let C be a central distribution of $\mathcal{Q}$ for $\{T_n\}_{n=1}^{\infty}$. Define

$$C^*(S_n, T_n) = R(S_n \mid T_n) C(T_n), \quad n \geq M.$$

If C^* is a probability measure, then it is a central distribution of $\mathcal{Q}$ for $\{(S_n, T_n)\}_{n=1}^{\infty}$.

Proof. By Theorem 9, for each $n \geq M$ we have

$$C_n(S_n, T_n) = R(S_n \mid T_n) C_n(T_n),$$

where $C_n(T_n)$ is a central distribution for T_n, and $C_n(S_n, T_n)$ is a central distribution for (S_n, T_n). Given that C^* is a probability measure, we have

$$\begin{aligned} \mathbb{D}_{\mathrm{KL}}\left[C_n(S_n, T_n) \,\|\, C^*(S_n, T_n)\right] &= \mathbb{D}_{\mathrm{KL}}\left[R(S_n \mid T_n) C_n(T_n) \,\|\, R(S_n \mid T_n) C(T_n)\right] \\ &= \mathbb{D}_{\mathrm{KL}}\left[C_n(T_n) \,\|\, C(T_n)\right]. \end{aligned}$$

Since C is a central distribution for $\{T_n\}_{n=1}^{\infty}$, by definition,

$$\lim_{n \to \infty} \mathbb{D}_{\mathrm{KL}}\left[C_n(T_n) \,\|\, C(T_n)\right] = 0.$$

It follows that

$$\lim_{n \to \infty} \mathbb{D}_{\mathrm{KL}}\left[C_n(S_n, T_n) \,\|\, C^*(S_n, T_n)\right] = 0,$$

and therefore C^* is a central distribution for $\{(S_n, T_n)\}_{n=1}^{\infty}$. □

Theorems 11 and 12 establish the conditions for asymptotic centrality; however, proving general existence and uniqueness for infinite sequences remains an open problem. As such, we present the following construction as a conjectured framework. Under standard regularity conditions, this approach motivates the functional form of central priors and serves as a conceptual bridge to stimulate future formal proofs.

Let $\mathcal{Q}$ be a family of probability distributions indexed by a parameter vector Θ. Suppose $\{T_n\}_{n=1}^{\infty}$ is a sequence of consistent estimators of Θ. Let $P(\theta \mid T_n)$ denote the posterior distribution obtained from an arbitrary smooth positive prior.

Under regular parametric conditions, posterior distributions based on consistent estimators become asymptotically insensitive to the choice of smooth positive prior.

Motivated by this phenomenon, we heuristically derive the asymptotic form of a central prior by considering the posterior density evaluated at the point $t = \theta$. A similar asymptotic construction also appears in [3].

Specifically, we define

$$c(\theta) \propto \lim_{n \to \infty} \frac{P(\theta \mid T_n = t)|_{t=\theta}}{\mu_n}, \tag{20}$$

where $\{\mu_n\}_{n=1}^{\infty}$ is a sequence of positive constants chosen so that the limit exists and the convergence is uniform over θ.

Let k denote the normalizing constant converting proportionality into equality. Since the logarithm is a continuous function, Equation 20 implies

$$\lim_{n\to\infty} \sup_{\theta} \left| \ln \frac{P(\theta \mid T_n = t)|_{t=\theta}}{c(\theta)} - \ln \mu_n + \ln k \right| = 0.$$

For every θ, since T_n is a consistent estimator, the sampling distribution $Q\left(T_n \mid \theta\right)$ becomes increasingly concentrated around θ as $n \to \infty$. Therefore, we may write

$$\lim_{n\to\infty} \sup_{\theta} \left| \sum_{t_n} Q(t_n \mid \theta) \ln \frac{P(\theta \mid t_n)}{c(\theta)} - \ln \mu_n + \ln k \right| = 0.$$

Furthermore, under the usual regularity conditions, the posterior distribution becomes asymptotically insensitive to the choice of smooth positive prior. Motivated by this asymptotic prior-insensitivity, we heuristically replace $P(\theta \mid T_n)$ with $c(\theta \mid T_n)$, obtaining

$$\lim_{n\to\infty} \sup_{\theta} \left| \sum_{t_n} Q(t_n \mid \theta) \ln \frac{Q(t_n \mid \theta)}{C(t_n)} - \ln \mu_n + \ln k \right| = 0.$$

Letting

$$\lambda_n = \ln \mu_n - \ln k,$$

we observe that

$$\lim_{n\to\infty} \sup_{\theta} \left| \mathbb{D}_{\mathrm{KL}} \left[Q\left(T_n \mid \theta\right) \| C(T_n) \right] - \lambda_n \right| = 0.$$

With appropriate additional assumptions, the conditions of Theorem 11 are satisfied. Thus, under the stated regularity assumptions and heuristic asymptotic arguments, $c(\theta)$ is expected to act as a central prior for the estimator sequence $\{T_n\}_{n=1}^{\infty}$.

A central distribution over an estimator sequence cannot by itself be used to perform inference at the level of observed outcomes. To do so, we must convert it into a distribution over the primary variables of interest. Theorems 9 and 12 provide a simple method for this conversion.

When dealing with layered ignorance and seeking the central distribution of a hierarchy of distributions, as defined in Definition 2.2, we typically encounter situations in which a given level of the hierarchy contains only finitely many alternatives. In such cases, and under appropriate regularity conditions, if there exists a consistent estimator for the index identifying these alternatives, then the central distribution reduces to an equally weighted average of the corresponding representative central distributions.

Example 2.1 (Binomial Experiment)**.** Consider a binomial experiment consisting of n independent Bernoulli trials. Let $\mathcal{Q}$ denote the family of probability distributions corresponding to different values of the success probability $\theta \in [0,1]$, with θ serving as the indexing parameter for $\mathcal{Q}$. Let T_n be the sample proportion of successes in the n trials. Since T_n is a consistent estimator of θ, we can apply Equation 20 to determine a central distribution for the sequence $\{T_n\}_{n=1}^{\infty}$.

Assuming a uniform prior on θ, Bayes' rule gives the posterior distribution

$$P\left(\theta \,\middle|\, T_n = \tfrac{k}{n}\right) = (n+1)\binom{n}{k}\theta^k(1-\theta)^{n-k},$$

which is a Beta$(k+1, n-k+1)$ distribution. Using Stirling's formula, we derive its asymptotic form:

$$P\left(\theta \,\middle|\, T_n = \tfrac{k}{n}\right) \sim \frac{(n+1)n^{(n+\frac{1}{2})}}{\sqrt{2\pi}(n-k)^{(n-k+\frac{1}{2})}k^{(k+\frac{1}{2})}}\,\theta^k(1-\theta)^{n-k}.$$

Evaluating the asymptotic expression at $\frac{k}{n} = \theta$, we obtain

$$P(\theta \mid T_n = t)|_{t=\theta} \sim \frac{n+1}{\sqrt{2\pi n}}\,\theta^{-\frac{1}{2}}(1-\theta)^{-\frac{1}{2}}.$$

Therefore, if we define

$$\mu_n = \frac{n+1}{\sqrt{2\pi n}},$$

the following limit exists and the convergence is uniform over $\theta \in (0,1)$:

$$\lim_{n\to\infty} \frac{P(\theta \mid T_n = t)|_{t=\theta}}{\mu_n} = \theta^{-\frac{1}{2}}(1-\theta)^{-\frac{1}{2}}.$$

Example 2.2 (Fused Multinomial Model)**.** Let $\{S_n\}_{n=1}^{\infty}$ be a sequence in which each $S_n = (X_1, X_2, \ldots, X_n)$ represents n independent trials of an experiment with k possible outcomes, and $\mathcal{Q}$ be a set of probability distributions over S_n indexed by a real vector θ. Let $f : \mathcal{X} \to \mathcal{Y}$ be a deterministic function that maps each outcome of X_i to a label $Y_i = f(X_i)$, where $\mathcal{Y} = \{1, \ldots, m\}$. Assume that for all i and $Q_\theta \in \mathcal{Q}$, if $f(x_i) = f(x_i')$ we have

$$Q_\theta(x_1, x_2, \ldots, x_n) = Q_\theta(x_1', x_2', \ldots, x_n').$$

That is, the conditional distribution $Q_\theta(X_i \mid Y_i = y)$ is uniform over the preimage $f^{-1}(y)$.

We can consider $\theta = (\theta_1, \ldots, \theta_m)$ to be a probability vector over $\mathcal{Y}$, where each θ_y specifies the probability of observing label y. For a sequence S_n, let n_y denote the number of times label $y \in \mathcal{Y}$ appears. Let $|f^{-1}(y)|$ denote the number of outcomes in $\mathcal{X}$ that are mapped to label y by f. Then the probability of observing S_n under $Q_\theta \in \mathcal{Q}$, denoted $Q(S_n \mid \theta)$, is:

$$Q(S_n \mid \theta) = \prod_{y=1}^{m} \left(\frac{1}{|f^{-1}(y)|} \cdot \theta_y \right)^{n_y}.$$

Now, consider the estimator sequence $T_n = \frac{1}{n}(n_1, n_2, \ldots, n_m)$, which is a consistent estimator for θ. The conditional distribution $Q_\theta(S_n \mid T_n)$ is then given by:

$$R(S_n \mid T_n) = \frac{1}{n!} \prod_{y=1}^{m} \frac{n_y!}{|f^{-1}(y)|^{n_y}}. \tag{21}$$

Note that this distribution is independent of θ. Therefore, by Theorem 12, a central distribution for S_n can be obtained from a central distribution for T_n.

As $n \to \infty$, the distribution of T_n converges to a multivariate Gaussian distribution. Using this fact and Equation 20, we can determine a central prior for $\{T_n\}_{n=1}^{\infty}$:

$$c(\theta) \propto \prod_{y=1}^{m} \theta_y^{-\frac{1}{2}}.$$

This is a Dirichlet prior and induces a Dirichlet-multinomial distribution for each T_n:

$$C(T_n) = \frac{\Gamma\left(\frac{m}{2}\right)\Gamma(n+1)}{\Gamma\left(n+\frac{m}{2}\right)} \prod_{y=1}^{m} \frac{\Gamma\left(n_y+\frac{1}{2}\right)}{\Gamma\left(\frac{1}{2}\right)\Gamma(n_y+1)}.$$

Note that C is a central distribution for the sequence $\{T_n\}_{n=1}^{\infty}$, viewed as a probability measure on the infinite sequence T_∞. The notation $C(T_n)$ denotes the marginal distribution of T_n.

By Theorem 12 and Equation 21, we can derive a central distribution for $\{S_n\}_{n=1}^{\infty}$, which induces the following distribution over each S_n:

$$C(S_n) = \frac{\Gamma\left(\frac{m}{2}\right)}{\Gamma\left(n+\frac{m}{2}\right)} \prod_{y=1}^{m} \frac{\Gamma\left(n_y+\frac{1}{2}\right)}{\Gamma\left(\frac{1}{2}\right)|f^{-1}(y)|^{n_y}}.$$

3 Illustrative Examples

We now present some examples of central distributions and demonstrate how they can be used for feature selection.

3.1 Attribute Selection

In this example, we consider a classification problem where the goal is to predict a target class, denoted by Y, from a set of observed nominal attributes $(A_1, A_2, \ldots, A_n)$. We assume that Y may depend on an unknown subset of the attributes, rather than the full set.

This reflects a form of sparsity in the relationship between the class and the attributes, where irrelevant features are ignored in the true generative model.

We can use the fused multinomial model of Example 2.2 to capture this relationship. As an example, assume that Y depends only on A_2, A_5. Then, we define the labeling function f as follows:

$$f(A_1, A_2, \ldots, A_n, Y) = (A_2, A_5, Y)\cdot$$

In a fused multinomial model with this labeling function, given A_2, A_5, the target class Y is conditionally independent from other attributes. Additionally,

$$P(A_1, A_2, \ldots, A_n \mid A_2, A_5) = P(A_1, A_3, A_4, A_6, \ldots, A_n)$$

is a uniform distribution.

For every possible subset of attributes, a distinct fused multinomial model of this form is constructed. Each such model corresponds to a set of plausible probability distributions. Collecting these sets yields our overall uncertainty representation, which forms a one-level hierarchy of distributions (Definition 2.2).

Let $C_{\mathcal{H}^*}$ denote a central distribution of the full hierarchy, and let C_{f_i} denote the central distribution of the fused multinomial model associated with the labeling function f_i. As discussed earlier, because the number of candidate models is finite, we have

$$C_{\mathcal{H}^*}(S_n) \propto \sum_i C_{f_i}(S_n).$$

To justify this result, observe that we can construct a consistent estimator for the model indicator I by examining equality relationships among the observed frequencies of outcomes. Specifically, the estimator identifies outcomes that occur with equal frequencies in the sample and returns the indicator corresponding to the model that supports those equalities.

As shown in Equation 1, the optimal Bayesian decision rule depends on the choice of loss function. In this example, we use the zero–one loss, under which the optimal strategy predicts the target class with the highest posterior probability. These posterior probabilities are computed using the central distribution derived in Example 2.2. In the present construction, we note that each subset-specific fused multinomial model is assigned its corresponding Jeffreys' prior. Consequently, the resulting predictive distribution can be interpreted as Bayesian model averaging over all candidate feature subsets, with each model equipped with its Jeffreys' prior. The central distribution framework provides a minimax-KL justification for this aggregation.

To see how this approach works in practice, we carried out a synthetic experiment with the following setup. We considered binary attributes and generated each attribute independently with equal probability for 0 and 1. The target class was defined as a deterministic function of a fixed subset of 3 out of 12 total attributes, as shown in Table 1. The same subset of relevant attributes and class assignment function were used across all trials.

Table 1: Mappings of 3-bit inputs under class functions Y_1 and Y_2

Input (bits)	Y_1(Input)	Y_2(Input)
000	a	A
001	b	B
010	c	A
011	d	C
100	e	D
101	f	D
110	g	D
111	h	E

For each experiment, we generated a training set of size 20 and a test set of size 60. A larger test set was used to reduce variability in performance estimates, taking advantage of the synthetic nature of the data. To assess robustness to irrelevant features, we progressively removed one attribute at a time from the full set of 12, starting with irrelevant ones and stopping just before removing any of the three relevant attributes.

At each step, we trained and tested two versions of the central distribution: one assuming a deterministic class assignment function (CD-Det), where the target class is a deterministic function of the relevant attributes, and one allowing for a conditional (probabilistic) class assignment distribution (CD-Prob). Both versions are constructed by aggregating over 2^{12} fused multinomial models, corresponding to all subsets of the 12 attributes. As a baseline, we also trained a decision tree classifier (DTree). All methods were trained on the same training data and evaluated on the same test data. Each configuration was repeated for 100 independent trials, and we reported the average accuracy (correct predictions divided by total test instances) over these trials.

Figure 1: Accuracy vs. Number of Attributes (Function Y_1, 20 train samples)

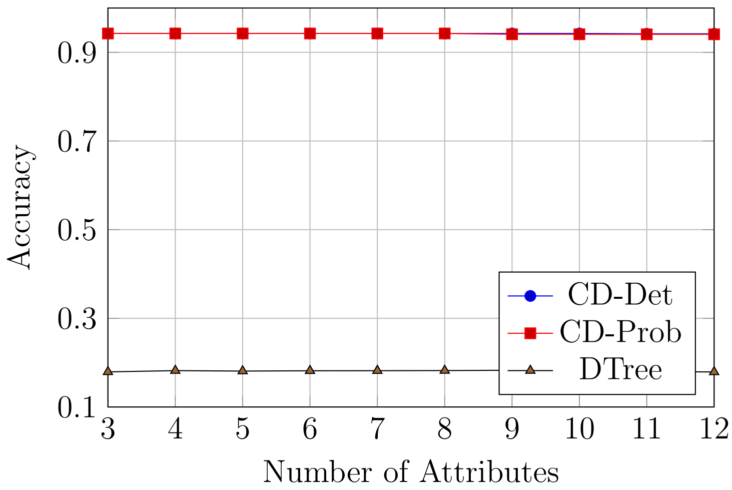
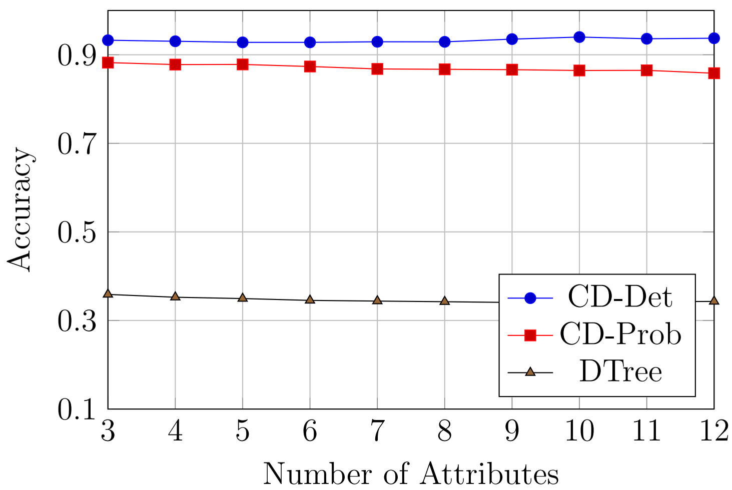
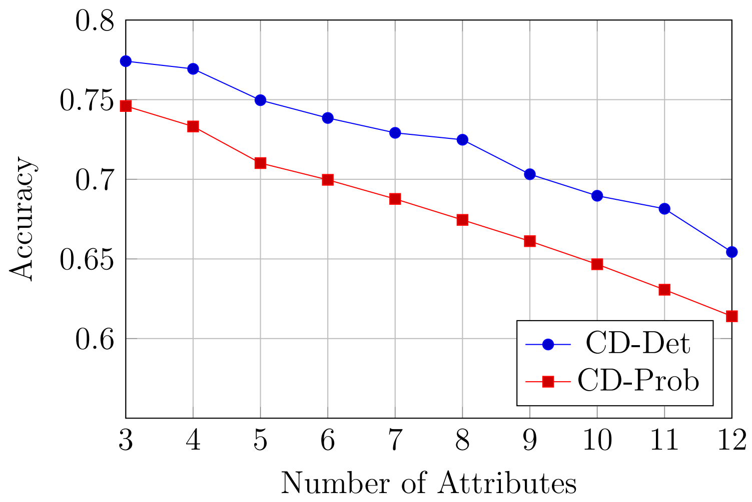


Figure 2: Accuracy vs. Number of Attributes (Function Y_2, 20 train samples)



The results, shown in Figures 1 and 2, reveal notable differences in performance depending on the underlying class assignment function. For function Y_1, both deterministic and probabilistic methods achieve high accuracy, with the deterministic version of the central distribution slightly outperforming the other. In contrast, function Y_2 is more difficult to predict, leading to a clearer separation in performance between the two methods. This difficulty primarily arises from strong non-deterministic dependencies between subsets of input bits and the assigned class. For instance, when the first input bit of Y_2 is 1, the class is often D, creating a misleading probabilistic association. In this case, the deterministic central distribution remains the most accurate overall, while the probabilistic version shows reduced performance due to misleading patterns.

Figure 3: Accuracy vs. Number of Attributes (Function Y_2, 10 train samples)



While removing irrelevant attributes generally led to slight improvements in accuracy, the effect was not substantial in our primary experiments with 20 training samples. However, we observed that the benefit of attribute removal becomes more noticeable when the training set is smaller. In particular, additional experiments with 10 training samples showed a clearer performance gain as irrelevant features were removed, indicating that the impact of irrelevant attributes is more pronounced in low-data regimes (see Figure 3).

4 Conclusion

We introduced the central distribution, a minimax-KL representative of a set, or more generally a hierarchy, of probability distributions. This provides a principled mechanism for converting partial or set-valued probabilistic knowledge into a single probability distribution suitable for Bayesian inference.

We showed that central distributions admit optimization-theoretic characterizations through convex-concave saddle point problems and established sufficient conditions for centrality based on the equalization of KL divergences. We further introduced central divergence as a measure of representational quality and derived bounds that enable the approximation of central distributions without requiring explicit knowledge of the exact solution.

For parametric families, we defined the notion of a central prior and established a sufficient fixed-point condition connecting central priors to Bernardo's reference prior methodology. This relationship suggests that central priors can recover reference priors in some settings and may extend naturally to more general forms of partial knowledge.

We also extended the framework to infinite sequences of random variables through the notion of asymptotic central distributions, providing a bridge between finite-sample central distributions and their asymptotic counterparts.

Finally, we argued that uncertainty about relevant feature subsets can naturally be expressed as a hierarchy of distributions. Within this framework, central distributions provide a mechanism for aggregating competing structural hypotheses and may induce a preference for simpler models when simpler dependency structures are compatible with a larger collection of hypotheses.

Several important questions remain open. In particular, existence and uniqueness results for general hierarchies, computational methods for large-scale problems, and a rigorous asymptotic theory for central priors require further investigation. We hope that the central distribution framework provides a useful foundation for studying objective Bayesian inference, hierarchical uncertainty, and feature selection from a unified information-theoretic perspective.

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A Expectation as a Linear Map

While studying linear algebra, one quickly encounters the idea that linear maps often reveal the essential structure of a mathematical theory. Probability theory provides a familiar example: expectation is a linear operation. This observation motivates a simple question: how much of elementary probability theory can be reconstructed if expectation, rather than probability, is taken as the starting point? In this note, we explore this question by treating expectation as a positive normalized linear functional on a space of actions.

Definition A.1 (Sample Space)**.** A sample space, denoted by $\mathcal{S}$, is the set of all possible outcomes of a random experiment. That is, the outcome of a random experiment can *always* be represented by *exactly one* element of $\mathcal{S}$.

Definition A.2 (Action Space)**.** Let $\mathcal{S}$ be a sample space. The action space of $\mathcal{S}$, denoted by $\mathcal{A}_{\mathcal{S}}$, is the set of all functions from $\mathcal{S}$ to $\mathbb{R}$, where $\mathbb{R}$ is the field of real numbers.

Proposition. The action space equals $\mathbb{R}^{\mathcal{S}}$, and is a vector space.

We can think of the members of $\mathcal{A}_{\mathcal{S}}$ as hypothetical actions or gambles. Each action associates an amount of loss with every possible outcome $s \in \mathcal{S}$. This loss is quantified by a real number. A negative loss indicates profit.

Definition A.3 (Sure Thing)**.** The *sure thing*, denoted by c, is an element of $\mathcal{A}_{\mathcal{S}}$, such that $c(s) = 1$ for every $s \in \mathcal{S}$.

Definition A.4. Let $\mathcal{E} \subseteq \mathcal{S}$. The restriction to $\mathcal{E}$ operator, denoted by $R_{\cap\mathcal{E}}$, is an operator on $\mathcal{A}_{\mathcal{S}}$, such that $\ell' = R_{\cap\mathcal{E}}(\ell_{\mathcal{S}})$ for any $\ell_{\mathcal{S}} \in \mathcal{A}_{\mathcal{S}}$ is an element of $\mathcal{A}_{\mathcal{S}}$ and

$$\ell'(s) = \begin{cases} \ell_{\mathcal{S}}(s) & s \in \mathcal{E}, \\ 0 & s \in \mathcal{S} - \mathcal{E}. \end{cases}$$

We denote the range of $R_{\cap\mathcal{E}}$ by $\mathcal{A}_{\mathcal{S}\cap\mathcal{E}}$.

Proposition. The restriction operator is a linear map.

Proposition. $\mathcal{A}_{\mathcal{S}\cap\mathcal{E}}$ is an invariant subspace of $\mathcal{A}_{\mathcal{S}}$ under $R_{\cap\mathcal{E}}$. If $\{\mathcal{E}_1, \mathcal{E}_2, \ldots\}$ is a partition of $\mathcal{S}$, then

$$\mathcal{A}_{\mathcal{S}} = \mathcal{A}_{\mathcal{S}\cap\mathcal{E}_1} \oplus \mathcal{A}_{\mathcal{S}\cap\mathcal{E}_2} \oplus \ldots,$$

where $\oplus$ indicates direct sum [1].

The restriction operator restricts an action to a subset of the sample space by simply ignoring its payoff for any outcome that is not in that subset.

Definition A.5 (Linear Functional)**.** Let $\mathcal{V}$ be a vector space over a field $\mathbb{F}$. A linear functional on $\mathcal{V}$ is a linear map from $\mathcal{V}$ to $\mathbb{F}$.

Definition A.6 (Expectation)**.** An *expectation* for $\mathcal{S}$ is a linear functional on $\mathcal{A}_{\mathcal{S}}$, denoted by $\mathbb{E}_{\mathcal{S}}$, with the following properties:

- For the sure thing $c_{\mathcal{S}}$, we have $\mathbb{E}_{\mathcal{S}}\left[c_{\mathcal{S}}\right] = 1$.
- For any $\mathcal{E} \subseteq \mathcal{S}$, we have $\mathbb{E}_{\mathcal{S}}\left[R_{\cap\mathcal{E}}(c_{\mathcal{S}})\right] \geq 0$.

Proposition. $\mathbb{E}_{\mathcal{S}}$ is an element of the dual space of $\mathcal{A}_{\mathcal{S}}$.

Definition A.7 (Conditional Expectation)**.** Let $\mathbb{E}_{\mathcal{S}}$ be an expectation for $\mathcal{S}$ and let $\mathcal{E} \subseteq \mathcal{S}$. $\mathbb{E}_{\mathcal{S}}$ conditioned on $\mathcal{E}$, denoted by $\mathbb{E}_{\mathcal{S}|\mathcal{E}}$, is a linear map on $\mathcal{S}$ with the following properties:

- For the sure thing $c_{\mathcal{S}}$, we have $\mathbb{E}_{\mathcal{S}|\mathcal{E}}\left[c_{\mathcal{S}}\right] = 1$.
- For any $\ell_{\mathcal{S}} \in \mathcal{A}_{\mathcal{S}}$, and $\mathcal{E} \subseteq \mathcal{S}$, we have $\mathbb{E}_{\mathcal{S}|\mathcal{E}}\left[\ell_{\mathcal{S}}\right] = 0$ if and only if $\mathbb{E}_{\mathcal{S}}\left[R_{\cap\mathcal{E}}(\ell_{\mathcal{S}})\right] = 0$.

Proposition. The conditional expectation $\mathbb{E}_{\mathcal{S}|\mathcal{E}}$ is only well-defined when $\mathbb{E}_{\mathcal{S}}\left[R_{\cap\mathcal{E}}(c_{\mathcal{S}})\right] \neq 0$.

It is important to note that conditioning on $\mathcal{E}$ fundamentally differs from restricting to $\mathcal{E}$. When we restrict an action, we simply ignore its payoffs outside of $\mathcal{E}$. When we condition on $\mathcal{E}$, however, we act with the certain knowledge that $\mathcal{E}$ has actually occurred. As $\mathcal{E}$ becomes increasingly unlikely, the disparity between these two concepts grows, because conditioning forces a rare event to become our new baseline. Yet, a fundamental exception occurs when our expected payoff within $\mathcal{E}$ is zero. No matter how unlikely $\mathcal{E}$ is, knowing that it has occurred cannot transform an expected payoff of zero into a non-zero value. Therefore, we require the null space of the conditional expectation to coincide perfectly with the null space induced by the restriction operator.

At this point, we have enough structure to reason about uncertainty and evaluate actions. Interestingly, we do not need to explicitly define probability. However, in order to construct a more familiar framework, we provide a definition of conventional probability.

Definition A.8 (Probability)**.** We define the probability of $\mathcal{E} \subseteq \mathcal{S}$ as follows:

$$P(\mathcal{E}) = \mathbb{E}_{\mathcal{S}}\left[R_{\cap\mathcal{E}}(c_{\mathcal{S}})\right] \cdot$$

Theorem 13. *Definition A.8 satisfies Kolmogorov's axioms of probability.*

We now establish the familiar relationship between conditional expectation and normal expectation.

Lemma 5. *Let $\mathcal{F}$ and $\mathcal{G}$ be two linear functionals defined on a vector space $\mathcal{V}$. $\mathcal{F}$ and $\mathcal{G}$ have the same null space if and only if $\mathcal{F} = \lambda\mathcal{G}$ for some $\lambda \neq 0$.*

Proof. If $\mathcal{F} = \lambda\mathcal{G}$, $\mathcal{F}[v] = 0$ if and only if $\mathcal{G}[v] = 0$ for every $v \in \mathcal{V}$. This proves the if part.

We prove the only if part by contradiction. Assume $\mathcal{F} \neq \lambda\mathcal{G}$. Then there exist $u, v \in \mathcal{V}$ such that

$$\mathcal{F}[u]\,\mathcal{G}[v] - \mathcal{F}[v]\,\mathcal{G}[u] \neq 0\cdot$$

Let $w = \mathcal{F}[u]\,v - \mathcal{F}[v]\,u$. Since $\mathcal{V}$ is a vector space $w \in \mathcal{V}$. $\mathcal{F}$ and $\mathcal{G}$ are linear. Hence, $\mathcal{F}[w] = 0$ and $\mathcal{G}[w] = \mathcal{F}[u]\,\mathcal{G}[v] - \mathcal{F}[v]\,\mathcal{G}[u] \neq 0$ and the null spaces of $\mathcal{F}$ and $\mathcal{G}$ are not equal. This contradiction completes the proof. □

Theorem 14. *Let $\ell_{\mathcal{S}} \in \mathcal{A}_{\mathcal{S}}$ and $\mathcal{E} \subseteq \mathcal{S}$, where $\mathcal{S}$ is a sample space. If $R_{\cap\mathcal{E}}$ is the restriction operator to $\mathcal{E}$, we have*

$$\mathbb{E}_{\mathcal{S}|\mathcal{E}}[\ell_{\mathcal{S}}] = \frac{\mathbb{E}_{\mathcal{S}}[R_{\cap\mathcal{E}}(\ell_{\mathcal{S}})]}{P(\mathcal{E})}\cdot$$

Proof. For an arbitrary $\ell \in \mathcal{A}_{\mathcal{S}}$, let $\ell' = \ell - R_{\cap\mathcal{E}}(\ell)$. Clearly, $R_{\cap\mathcal{E}}(\ell')$ is the zero function, and therefore $\mathbb{E}_{\mathcal{S}}[R_{\cap\mathcal{E}}(\ell')] = 0$. Since $\mathcal{A}_{\mathcal{S}}$ is a vector space we have $\ell' \in \mathcal{A}_{\mathcal{S}}$, and the definition of conditional expectation implies $\mathbb{E}_{\mathcal{S}|\mathcal{E}}[\ell'] = 0$. Therefore

$$\mathbb{E}_{\mathcal{S}|\mathcal{E}}[\ell] - \mathbb{E}_{\mathcal{S}|\mathcal{E}}[R_{\cap\mathcal{E}}(\ell)] = 0\cdot \tag{22}$$

Let $\mathcal{A}_{\mathcal{S}\cap\mathcal{E}}$ represent the range of $R_{\cap\mathcal{E}}$. $\mathcal{A}_{\mathcal{S}\cap\mathcal{E}}$ is a subspace of $\mathcal{A}_{\mathcal{S}}$. By the definition of conditional expectation, functionals obtained by the restrictions of $\mathbb{E}_{\mathcal{S}}$ and $\mathbb{E}_{\mathcal{S}|\mathcal{E}}$ to $\mathcal{A}_{\mathcal{S}\cap\mathcal{E}}$ have the same null space. That implies by Lemma 5 for every $\ell \in \mathcal{A}_{\mathcal{S}}$ we have

$$\mathbb{E}_{\mathcal{S}|\mathcal{E}}[R_{\cap\mathcal{E}}(\ell)] = \lambda\mathbb{E}_{\mathcal{S}}[R_{\cap\mathcal{E}}(\ell)]\cdot$$

Thus, by Equation 22

$$\mathbb{E}_{\mathcal{S}|\mathcal{E}}[\ell] = \lambda\mathbb{E}_{\mathcal{S}}[R_{\cap\mathcal{E}}(\ell)]\cdot$$

Provided that $\mathbb{E}_{\mathcal{S}|\mathcal{E}}$ is well-defined, for the sure thing $c_{\mathcal{S}}$ we have

$$\lambda = \frac{\mathbb{E}_{\mathcal{S}|\mathcal{E}}[c_{\mathcal{S}}]}{\mathbb{E}_{\mathcal{S}}[R_{\cap\mathcal{E}}(c_{\mathcal{S}})]} = \frac{1}{P(\mathcal{E})}\cdot$$

That proves the theorem. □

Definition A.9 (Dual Map)**.** Let $\mathcal{A}$ and $\mathcal{B}$ be two sets. Suppose $M : \mathcal{A} \to \mathcal{B}$ is a map from $\mathcal{A}$ to $\mathcal{B}$. The dual map of M is the map $M^* : \mathbb{R}^{\mathcal{B}} \to \mathbb{R}^{\mathcal{A}}$, defined for each $f \in \mathbb{R}^{\mathcal{B}}$ by

$$M^*(f) = f \circ M\cdot$$

Proposition. The dual map is a linear map.

Theorem 15. *Let $\mathcal{S}$ be a sample space. Suppose $M : \mathcal{S} \to \mathcal{S}'$ is a map from $\mathcal{S}$ to another set $\mathcal{S}'$. If $\mathbb{E}_{\mathcal{S}}$ is an expectation for $\mathcal{S}$, then $\mathbb{E}_{\mathcal{S}'} = M^{**}(\mathbb{E}_{\mathcal{S}})$ is an expectation for $\mathcal{S}'$, where M^{**} is the dual map of M^*, and M^* is the dual map of M.*

Proof. M^{**} is a dual map and therefore is linear.

By definition for every $\ell_{\mathcal{S}'} \in \mathcal{A}_{\mathcal{S}'}$ we have

$$\mathbb{E}_{\mathcal{S}'}\left[\ell_{\mathcal{S}'}\right] = \mathbb{E}_{\mathcal{S}} \circ \ell_{\mathcal{S}'} \circ M\cdot$$

We can easily verify that

$$c_{\mathcal{S}'} \circ M = c_{\mathcal{S}}\cdot$$

Therefore, $\mathbb{E}_{\mathcal{S}'}\left[c_{\mathcal{S}'}\right] = 1$.

On the other hand for any $\mathcal{E}' \subseteq \mathcal{S}'$, we have

$$R_{\cap\mathcal{E}'}(c_{\mathcal{S}'}) \circ M = R_{\cap\mathcal{E}}(c_{\mathcal{S}}),$$

where

$$\mathcal{E} = \{s \in \mathcal{S} : M(s) \in \mathcal{E}'\}\cdot$$

This implies $\mathbb{E}_{\mathcal{S}'}\left[R_{\cap\mathcal{E}'}(c_{\mathcal{S}'})\right] \geq 0$. $\square$

B Latent Regression Model

We consider C classes (or datasets), indexed by $c = 1, \dots, C$. For each class c, we observe a data matrix $X^{(c)} \in \mathbb{R}^{N_c \times M}$, where each row $\boldsymbol{x}_i^{(c)} \in \mathbb{R}^M$ is associated with a D-dimensional latent variable $\boldsymbol{z}_i^{(c)} \in \mathbb{R}^D$.

Thus, each class is associated with a collection of latent variables

$$\boldsymbol{z}^{(c)} = \{\boldsymbol{z}_i^{(c)}\}_{i=1}^{N_c}, \qquad \boldsymbol{z}_i^{(c)} \in \mathbb{R}^D.$$

For each class c, the latent variables are assumed independent and identically distributed according to

$$\boldsymbol{z}_i^{(c)} \sim \mathcal{N}\big(\boldsymbol{\mu}^{(c)}, \tau_c^{-1} I_D\big), \qquad c = 2, \dots, C,$$

where $\boldsymbol{\mu}^{(c)} \in \mathbb{R}^D$ is the class-specific latent mean and $\tau_c > 0$ is a precision parameter controlling the latent dispersion.

To remove redundancy, we fix the latent distribution of the first class as a reference coordinate system. Specifically, we impose

$$\boldsymbol{\mu}^{(1)} = \mathbf{0}, \qquad \tau_1 = 1,$$

so that

$$\boldsymbol{z}_i^{(1)} \sim \mathcal{N}(\mathbf{0}, I_D).$$

This anchoring removes global translation and scaling symmetries without restricting the expressive capacity of the model. The latent geometry of each class is therefore interpreted relative to the reference class $c = 1$.

Let $\phi : \mathbb{R}^D \to \mathbb{R}^K$ denote a fixed nonlinear basis mapping. The design matrix of class c is defined as $Z^{(c)} \in \mathbb{R}^{N_c \times K}$ with entries

$$Z_{ik}^{(c)} = \phi_k(\boldsymbol{z}_i^{(c)}).$$

We collect the decoder weights in a shared matrix

$$W = [\boldsymbol{w}_1 \ \dots \ \boldsymbol{w}_M] \in \mathbb{R}^{K \times M},$$

where each column $\boldsymbol{w}_j \in \mathbb{R}^K$ corresponds to the weights for the j-th observed dimension across all classes.

Conditioned on $\boldsymbol{z}^{(c)}$ and W, observations are generated as

$$p(X^{(c)} \mid \boldsymbol{z}^{(c)}, W) = \prod_{i=1}^{N_c} \prod_{j=1}^{M} \mathcal{N}\left(x_{ij}^{(c)} \;\middle|\; \phi(\boldsymbol{z}_i^{(c)})^\top \boldsymbol{w}_j, \beta_j^{-1}\right),$$

or equivalently,

$$\boldsymbol{x}_j^{(c)} \sim \mathcal{N}\big(Z^{(c)}\boldsymbol{w}_j, \beta_j^{-1} I_{N_c}\big)\,,$$

where $\boldsymbol{x}_j^{(c)}$ denotes the j-th column of $X^{(c)}$ and $\beta_j > 0$ is the observation precision of that column.

To regularize the decoder, we assume noninformative priors for the latent parameters $\{\boldsymbol{\mu}^{(c)}, \tau_c\}_{c=2}^{C}$, and place independent Gaussian priors on the columns of W:

$$p(W) = \prod_{j=1}^{M} \mathcal{N}\left(\boldsymbol{w}_j \;\middle|\; \mathbf{0}, \alpha_j^{-1} I_K\right).$$

The full joint distribution factorizes as

$$p(W)\prod_{c=1}^{C} p(X^{(c)} \mid \boldsymbol{z}^{(c)}, W)p(\boldsymbol{z}^{(c)} \mid \boldsymbol{\mu}^{(c)}, \tau_c)p(\boldsymbol{\mu}^{(c)}, \tau_c).$$

This defines a shared nonlinear generative model in which all classes are coupled through the common decoder W, while retaining class-specific latent geometry via $(\boldsymbol{\mu}^{(c)}, \tau_c)$.

B.1 Variational Inference for the Single-Class Case

To simplify the exposition, we now consider the single-class case. Since only one class is present, we remove the superscript (c) from the notation for clarity.

Furthermore, we restrict the latent variable to be scalar,

$$\boldsymbol{z} = \{z_i\}_{i=1}^{N} \qquad z_i \in \mathbb{R},$$

so that

$$p(\boldsymbol{z} \mid \mu, \tau) = \prod_{i=1}^{N} \mathcal{N}(z_i \mid \mu, \tau^{-1}).$$

We treat μ and τ as unknown parameters to be inferred, using the standard joint Jeffreys prior for the mean and precision of a Gaussian model,

$$p(\mu, \tau) \propto \tau^{-1}.$$

The nonlinear basis mapping reduces to

$$\phi : \mathbb{R} \to \mathbb{R}^K,$$

and the design matrix $Z \in \mathbb{R}^{N \times K}$ is defined by

$$Z_{ik} = \phi_k(z_i).$$

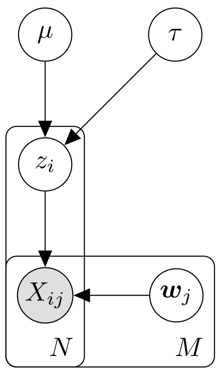

Figure 4: One class scalar latent regression model

We see the Bayesian network of this model in Figure 4. Exact posterior inference in this model is intractable due to the nonlinear dependence of the likelihood on $\boldsymbol{z}$. We therefore proceed by introducing a variational approximation.

In order to formulate a variational treatment of this model, we consider a variational distribution which factorizes between the latent variables and the parameters so that

$$q(\boldsymbol{z}, W, \mu, \tau) = q(\boldsymbol{z})q(W, \mu, \tau).$$

First, let us consider the derivation of the update equation for the factor $q(W, \mu, \tau)$. The log of the optimized factor is given by

$$\ln q^*(W, \mu, \tau) = \mathbb{E}_{q(\boldsymbol{z})} \left[\ln p(W, \mu, \tau \mid \boldsymbol{z}, X)\right] + \text{const}.$$

As can be seen from Figure 4, the d-separation criterion implies that

$$\mathbb{E}_{q(\boldsymbol{z})} \left[\ln p(W, \mu, \tau \mid \boldsymbol{z}, X)\right] = \mathbb{E}_{q(\boldsymbol{z})} \left[\ln p(\mu, \tau \mid \boldsymbol{z})\right] + \sum_j \mathbb{E}_{q(\boldsymbol{z})} \left[\ln p(\boldsymbol{w}_j \mid \boldsymbol{z}, X)\right].$$

Therefore, we have the following induced factorization for $q(W, \mu, \tau)$:

$$q(W, \mu, \tau) = q(\mu, \tau) \prod_{j=1}^{M} q(\boldsymbol{w}_j).$$

We obtain $q(\boldsymbol{w}_j)$ using the fact that each $\boldsymbol{w}_j$ is a weight vector of a Bayesian linear regression model:

$$\begin{aligned}\ln p(\boldsymbol{w}_j \mid \boldsymbol{z}, X) &= -\frac{\beta_j}{2} \|Z\boldsymbol{w}_j - \boldsymbol{x}_j\|^2 - \frac{\alpha_j}{2}\boldsymbol{w}_j^\top \boldsymbol{w}_j + \text{const} \\ &= -\frac{1}{2}\boldsymbol{w}_j^\top(\beta_j Z^\top Z + \alpha_j I)\boldsymbol{w}_j + \beta_j \boldsymbol{x}_j^\top Z\boldsymbol{w}_j + \text{const}.\end{aligned}$$

Taking the expectation with respect to $q(\boldsymbol{z})$, we obtain

$$\ln q^*(\boldsymbol{w}_j) = -\frac{1}{2}\boldsymbol{w}_j^\top(\beta_j\mathbb{E}\left[Z^\top Z\right] + \alpha_j I)\boldsymbol{w}_j + \beta_j\boldsymbol{w}_j^\top\mathbb{E}\left[Z\right]^\top \boldsymbol{x}_j + \text{const},$$

where expectations of matrices are taken element-wise. The resulting expression is quadratic in $\boldsymbol{w}_j$. Completing the square yields

$$q^*(\boldsymbol{w}_j) = \mathcal{N}(\boldsymbol{w}_j \mid \boldsymbol{\mu}_j, \Sigma_j).$$

The corresponding moments are

$$\begin{aligned}\mathbb{E}\left[\boldsymbol{w}_j\right] &= \boldsymbol{\mu}_j,\\ \mathbb{E}\left[\boldsymbol{w}_j\boldsymbol{w}_j^\top\right] &= \boldsymbol{\mu}_j\boldsymbol{\mu}_j^\top + \Sigma_j,\end{aligned}$$

with

$$\begin{aligned}\Sigma_j^{-1} &= \beta_j\mathbb{E}\left[Z^\top Z\right] + \alpha_j I,\\ \boldsymbol{\mu}_j &= \beta_j\Sigma_j\mathbb{E}\left[Z\right]^\top \boldsymbol{x}_j.\end{aligned}$$

Since the design matrix satisfies $Z_{ik} = \phi_k(z_i)$, we have

$$\mathbb{E}\left[Z^\top Z\right] = \sum_{i=1}^{N}\mathbb{E}\left[\phi(z_i)\phi(z_i)^\top\right].$$

Collecting the posterior means into a matrix gives

$$\mathbb{E}\left[W\right] = [\boldsymbol{\mu}_1 \ \ \dots \ \ \boldsymbol{\mu}_M].$$

Moreover, to simplify later expressions, define the diagonal precision matrix

$$B = \operatorname{diag}(\beta_1, \dots, \beta_M).$$

Then, we have

$$\sum_{j=1}^{M}\beta_j\mathbb{E}\left[\boldsymbol{w}_j\boldsymbol{w}_j^\top\right] = \mathbb{E}\left[W\right]B\mathbb{E}\left[W\right]^\top + \sum_{j=1}^{M}\beta_j\Sigma_j.$$

This quantity will appear in the update for the latent variables. Note that if $\alpha_j = 0$ for all j, the mean simplifies to

$$\mathbb{E}\left[W\right] = \mathbb{E}\left[Z^\top Z\right]^{-1}\mathbb{E}\left[Z\right]^\top X.$$

We now consider the update equation for the factor $q(\mu, \tau)$:

$$\begin{aligned}\ln q^*(\mu, \tau) &= \mathbb{E}_{q(\boldsymbol{z})}\left[\ln p(\mu, \tau \mid \boldsymbol{z})\right] + \text{const} \\ &= -\frac{\tau}{2}\left(\sum_i \mathbb{E}\left[z_i^2\right] - 2\mu \sum_i \mathbb{E}\left[z_i\right] + N\mu^2\right) \\ &\quad + \left(\frac{N}{2} - 1\right)\ln \tau + \text{const.}\end{aligned}$$

We define

$$\begin{aligned}\overline{z} &= \frac{1}{N}\sum_i \mathbb{E}\left[z_i\right], \\ \overline{zz} &= \frac{1}{N}\sum_i \mathbb{E}\left[z_i^2\right].\end{aligned}$$

We observe that $q^*(\mu, \tau)$, for $N > 1$ follows a Normal–Inverse–Gamma distribution:

$$q^*(\mu, \tau) = \mathcal{N}\left(\mu \;\middle|\; \overline{z}, \frac{1}{N\tau}\right)\text{Gamma}\left(\tau \;\middle|\; \frac{N-1}{2}, \frac{N}{2}(\overline{zz} - \overline{z}^2)\right).$$

This implies

$$\begin{aligned}\mathbb{E}\left[\tau\right] &= \frac{(N-1)}{N(\overline{zz} - \overline{z}^2)}, \\ \mathbb{E}\left[\mu\tau\right] &= \overline{z}\mathbb{E}\left[\tau\right].\end{aligned}$$

Finally, we derive the update equation for $q(\boldsymbol{z})$:

$$\begin{aligned}\ln q^*(\boldsymbol{z}) &= \mathbb{E}_{q(W,\mu,\tau)}\left[\ln p(X \mid \boldsymbol{z}, W) + \ln p(\boldsymbol{z} \mid \mu, \tau)\right] + \text{const} \\ &= \mathbb{E}_{q(W)}\left[\ln p(X \mid \boldsymbol{z}, W)\right] + \mathbb{E}_{q(\mu,\tau)}\left[\ln p(\boldsymbol{z} \mid \mu, \tau)\right] + \text{const.}\end{aligned}$$

We expand the terms $\ln p(X \mid \boldsymbol{z}, W)$ and $\ln p(\boldsymbol{z} \mid \mu, \tau)$. All expressions that are independent of $\boldsymbol{z}$ are treated as constants.

$$\begin{aligned}
\ln p(X \mid \boldsymbol{z}, W) &= \sum_j \ln \mathcal{N}(\boldsymbol{x}_j \mid Z\boldsymbol{w}_j, \beta_j^{-1} I) \\
&= -\frac{1}{2} \sum_j \beta_j \left\| Z\boldsymbol{w}_j - \boldsymbol{x}_j \right\|^2 + \text{const} \\
&= \sum_i \left(-\frac{1}{2} \phi(z_i)^\top \sum_j \beta_j \boldsymbol{w}_j \boldsymbol{w}_j^\top \phi(z_i) + \phi(z_i)^\top \sum_j \beta_j X_{ij} \boldsymbol{w}_j \right) + \text{const.}
\end{aligned}$$

$$\begin{aligned}
\ln p(\boldsymbol{z} \mid \mu, \tau) &= -\frac{\tau}{2} \sum_i (z_i - \mu)^2 + \text{const} \\
&= \sum_i \left(-\frac{\tau}{2} z_i^2 + \tau \mu z_i \right) + \text{const.}
\end{aligned}$$

Substituting these expansions into the variational update expression and taking the expectation yields:

$$\ln q^*(\boldsymbol{z}) = \sum_i \ln q^*(z_i),$$

where

$$\begin{aligned}
\ln q^*(z_i) = & \left(-\frac{1}{2} \phi(z_i)^\top \sum_j \beta_j \mathbb{E}\left[\boldsymbol{w}_j \boldsymbol{w}_j^\top \right] \phi(z_i) + \phi(z_i)^\top \mathbb{E}\left[W \right] B X_{i,:}^\top \right) \\
& + \left(-\frac{1}{2} \mathbb{E}\left[\tau \right] z_i^2 + \mathbb{E}\left[\tau \mu \right] z_i \right) + \text{const.}
\end{aligned}$$

Here $X_{i,:}$ denotes the i-th row of the observation matrix X.

Because the update expression separates as a sum over i, the optimal factor $q(\boldsymbol{z})$ factorizes as $\prod_i q(z_i)$. Define

$$\begin{aligned}
S^{-1} &= \sum_j \beta_j \mathbb{E}\left[\boldsymbol{w}_j \boldsymbol{w}_j^\top \right], \\
\boldsymbol{m}_i &= S \mathbb{E}\left[W \right] B X_{i,:}^\top.
\end{aligned}$$

Then, up to normalization,

$$q^*(z_i) \propto \mathcal{N}\left(\phi(z_i) \mid \boldsymbol{m}_i, S \right) \mathcal{N}\left(z_i \;\middle|\; \mathbb{E}\left[\mu \right], \mathbb{E}\left[\tau \right]^{-1} \right).$$

Note that since the basis mapping is nonlinear, this does not generally imply that $q^*(z_i)$ is Gaussian in z_i.

As seen previously, the update equations require expectations of the form

$$\mathbb{E}\left[\phi_i(z_k)\phi_j(z_k)\right], \quad \mathbb{E}\left[\phi_i(z_k)\right], \quad \mathbb{E}\left[z_k\right], \quad \mathbb{E}\left[z_k^2\right],$$

for all relevant indices i, j, k. These expectations typically do not admit closed forms and must be approximated numerically.